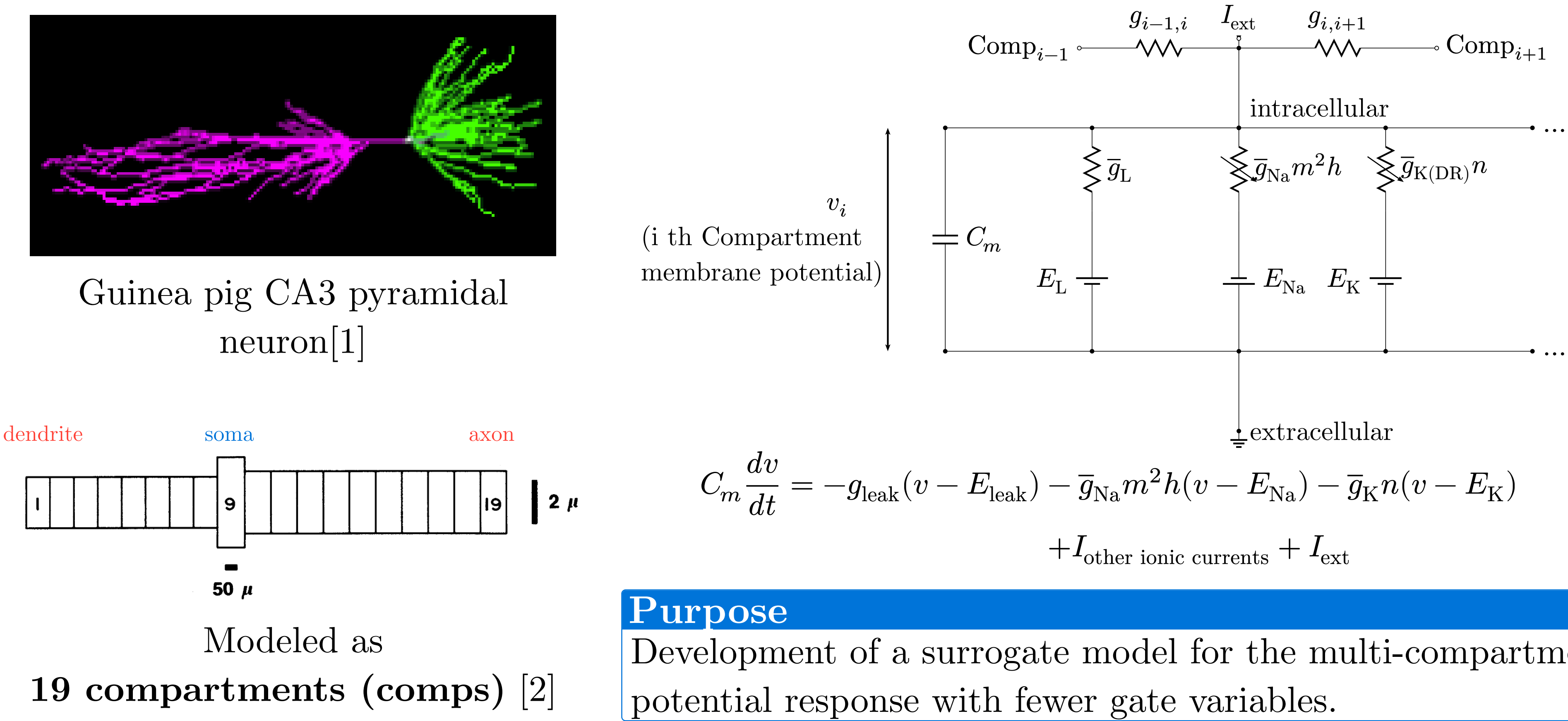


Introduction



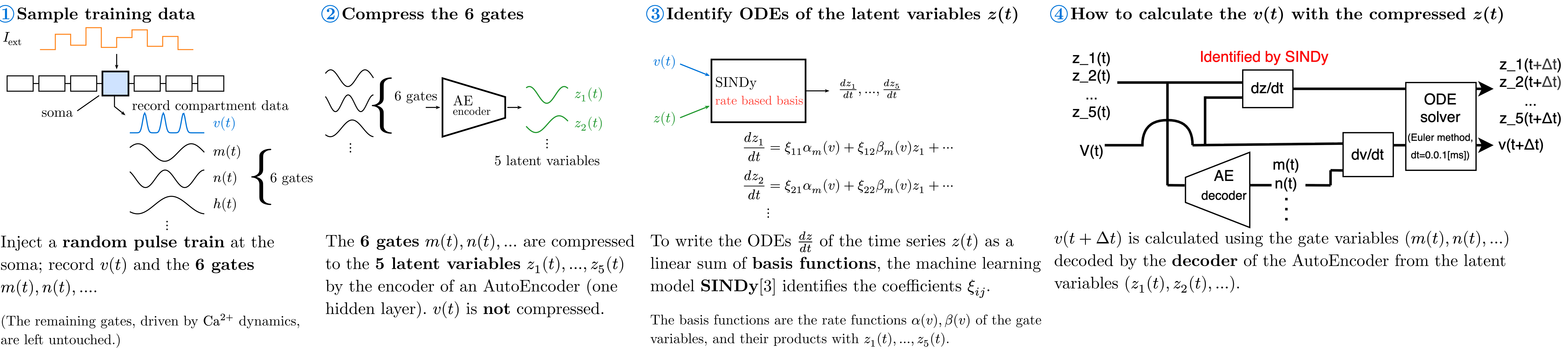
The conductances depend on **gate variables**, which follow ODEs with the rate functions α, β :

$$I_{\text{Na}} = \bar{g}_{\text{Na}} m^2 h (v - E_{\text{Na}})$$
$$\frac{dm}{dt} = \alpha_m(v)(1 - m) - \beta_m(v)m$$
$$\alpha_m(v) = 0.32(13.1 - v)/(\exp((13.1 - v)/4) - 1)$$
$$\beta_m(v) = 0.28(v - 40.1)/(\exp((v - 40.1)/5) - 1)$$

→ **10 gates and the potential v per comp**
→ **190 gates** for the 19 comps;
In large scale network simulations, the **large number of gate variables** becomes a **memory bottleneck**.

Purpose
Development of a surrogate model for the multi-compartment model, capable of reproducing the membrane potential response with fewer gate variables.

Methods



Results and Discussion

